

VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
UNIVERSITY OF SCIENCE
FACULTY OF MATHEMATICS AND COMPUTER SCIENCE



MIDTERM ESSAY · MEASURE THEORY AND INTEGRATION

COMPLETION OF MEASURE SPACES

Major: Mathematics

Specialisation: Analysis

Lecturer: Assoc. Prof. Bui Le Trong Thanh

Student: Huy **ID:** 24110022

Ho Chi Minh City, March 25, 2026

Measure Theory and Integration

Midterm Essay

Completion of Measure Spaces

A self-contained exposition following Yeh (2014)

Major	Mathematics
Specialisation	Analysis
Supervisor	Assoc. Prof. Bui Le Trong Thanh
Author	Huy ID 24110022

Ho Chi Minh City · 2026

Contents

Preliminaries	1
σ -algebras and measure spaces	1
Null sets and completeness	4
Completion of Measure Spaces	6
The completion construction	6
Uniqueness of the completion	9
From Borel to Lebesgue measure	11
References	14